



CSE3023 WEB APPLICATION DEVELOPMENT

SEMESTER 2 SESSION 2025/2026

**BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH
HONOURS**

LAB REPORT 6

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DATE OF SUBMISSION : 17 MEI 2026

TASK 1

```
<%--
Document   : SampleInsertionRecord
Created on : 17 May 2026, 9:12:42pm
Author    : Acer
--%>

<%@ page import="java.sql.*" %>

<html>
<head>
<title>Sample Insertion Record</title>
</head>
<body>

<h1>Lab 6 - Task 1 - Sample Insertion records into MySQL through JSP page</h1>

<%
    Connection conn = null;
    Statement stmt = null;

    try {

        Class.forName("com.mysql.cj.jdbc.Driver");

        conn = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/CSA3203",
            "root",
            "admin"
        );

        stmt = conn.createStatement();

        String sql = "INSERT INTO FirstTable VALUES ('Welcome to access MySQL database with JSP...!')";

        stmt.executeUpdate(sql);

        out.println("Record inserted successfully!");

    } catch (Exception e) {

        out.println("Error: " + e);

    }

%>

</body>
</html>
```

```
<!--
Document   : insertAuthor
Created on : 17 May 2026, 9:29:04 pm
Author    : Acer
-->
```

```
<html>
  <head>
    <title>Insert Author</title>
  </head>
  <body>

    <h1>Author Registration Form</h1>

    <form action="processAuthor.jsp" method="post">

      <table border="1" cellpadding="5">

        <tr>
          <td>Author No</td>
          <td>
            <input type="text" name="authno">
          </td>
        </tr>

        <tr>
          <td>Name</td>
          <td>
            <input type="text" name="name">
          </td>
        </tr>

        <tr>
          <td>Address</td>
          <td>
            <input type="text" name="address">
          </td>
        </tr>
      </table>
    </form>
  </body>
</html>
```

```
39 </tr>
40 <tr>
41 <td>City</td>
42 <td>
43 <input type="text" name="city">
44 </td>
45 </tr>
46
47 <tr>
48 <td>State</td>
49 <td>
50 <input type="text" name="state">
51 </td>
52 </tr>
53
54 <tr>
55 <td>Zip</td>
56 <td>
57 <select name="zip">
58 <option value="21000">21000</option>
59 <option value="22000">22000</option>
60 <option value="23000">23000</option>
61 <option value="24000">24000</option>
62 </select>
63 </td>
64 </tr>
65 <tr>
66 <td colspan="2">
67 <input type="submit" value="Submit">
68 <input type="reset" value="Cancel">
69 </td>
70 </tr>
71 </table>
72 </form>
73 </body>
74 </html>
```

```

1  <%--
2      Document      : processAuthor
3      Created on   : 17 May 2026, 9:30:38 pm
4      Author      : Acer
5  --%>
6
7  <%@page import="java.sql.*"%>
8  <%@page import="lab6.com.Author"%>
9
10 <!DOCTYPE html>
11 <html>
12 <head>
13     <title>Process Author</title>
14 </head>
15 <body>
16
17 <jsp:useBean id="author" class="lab6.com.Author" scope="request"/>
18
19 <jsp:setProperty name="author" property="*" />
20
21 <%
22
23 Connection conn = null;
24 PreparedStatement ps = null;
25
26 try {
27
28     Class.forName("com.mysql.cj.jdbc.Driver");
29
30     conn = DriverManager.getConnection(
31         "jdbc:mysql://localhost:3306/CSA3203",
32         "root",
33         "admin"
34     );
35
36     String sql = "INSERT INTO Author VALUES (?, ?, ?, ?, ?, ?)";

```

```

    ps = conn.prepareStatement(sql);

    ps.setString(1, author.getAuthno());
    ps.setString(2, author.getName());
    ps.setString(3, author.getAddress());
    ps.setString(4, author.getCity());
    ps.setString(5, author.getState());
    ps.setString(6, author.getZip());

    ps.executeUpdate();

    out.println("Author record inserted successfully!");
} catch (Exception e) {

    out.println("Error: " + e);

} finally {

    if(ps != null)
        ps.close();

    if(conn != null)
        conn.close();
}
%>

</body>
</html>

```

Databases

Create database

CSA3203

utf8mb4_0900_ai_ci

Create

☐ Check all

Create new table

Table name

Number of columns

Table structure
Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	message	varchar(100)	utf8mb4_0900_ai_ci		No	None			Change Drop

Lab 6 - Task 1 - Sample Insertion records into MySQL through JSP page

Record inserted successfully!

Reflection

1. How do you want to submit specific information from one form to next form?

Information can be submitted from one form to another using HTML form elements and JSP request methods such as `request.getParameter()`.

2. What happened if the field name you in `request.getParameter("field_name")` in second page is different from the field name you defined in first page?

If the field name in `request.getParameter()` is different from the form field name, the value cannot be retrieved, and it returns null.

TASK 2

```
Run SQL query/queries on database csa3203: ?
```

```
1 CREATE TABLE Author (  
2     authno VARCHAR(15) PRIMARY KEY,  
3     name VARCHAR(40),  
4     address VARCHAR(40),  
5     city VARCHAR(40),  
6     state VARCHAR(40),  
7     zip VARCHAR(40)  
8 );|
```

Author Registration Form

Author No	
Name	
Address	
City	
State	
Zip	21000 ▼
Submit Cancel	

Author Registration Form

Author No	s76025
Name	izza
Address	TAMAN SIMPANG BAKTI
City	TAIPING
State	PERAK
Zip	22000 ▼
Submit Cancel	

Author record inserted successfully!

```
/**
 *
 * @author Acer
 */
package lab6.com;

public class Author {

    private String authno;
    private String name;
    private String address;
    private String city;
    private String state;
    private String zip;

    public String getAuthno() {
        return authno;
    }

    public void setAuthno(String authno) {
        this.authno = authno;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getAddress() {
        return address;
    }
}
```

```
[-] public void setAddress(String address) {  
    [-]     this.address = address;  
    [-] }  
  
[-] public String getCity() {  
    [-]     return city;  
    [-] }  
  
[-] public void setCity(String city) {  
    [-]     this.city = city;  
    [-] }  
  
[-] public String getState() {  
    [-]     return state;  
    [-] }  
  
[-] public void setState(String state) {  
    [-]     this.state = state;  
    [-] }  
  
[-] public String getZip() {  
    [-]     return zip;  
    [-] }  
  
[-] public void setZip(String zip) {  
    [-]     this.zip = zip;  
    [-] }  
}
```

Reflection

1. What have you learnt from this exercise?

I learned how to connect JSP with MySQL database and insert records using PreparedStatement and JavaBeans.

2. Define step by step before you successfully perform the transaction in a database

- Create database
- Create table
- Create JSP form
- Create JavaBean
- Connect database using JDBC
- Execute SQL query
- Display result
- Close connection

TASK 3

```
1 CREATE TABLE Student (  
2     stuid VARCHAR(15) PRIMARY KEY,  
3     stuname VARCHAR(50),  
4     stuprogram VARCHAR(40),  
5     address VARCHAR(40)  
6 );
```

Student Registration Form

Student ID	UK12489
Student Name	Ahmad Salam
Program	BSc with IM ▼
Address	
Submit Cancel	

Student Registration Form

Student ID	UK12489
Student Name	izza
Program	BSc Soft. Eng. ▼
Address	taiping
Submit Cancel	

Process Student Registration

Student record inserted successfully!

Student ID	Name	Program	Address
UK12489	Ahmad Salam	BSc with IM	

Process Student Registration

Student record inserted successfully!

Student ID	Name	Program	Address
UK12489	izza	BSc Soft. Eng.	taiping

Student Registration Form

Student ID

Student Name

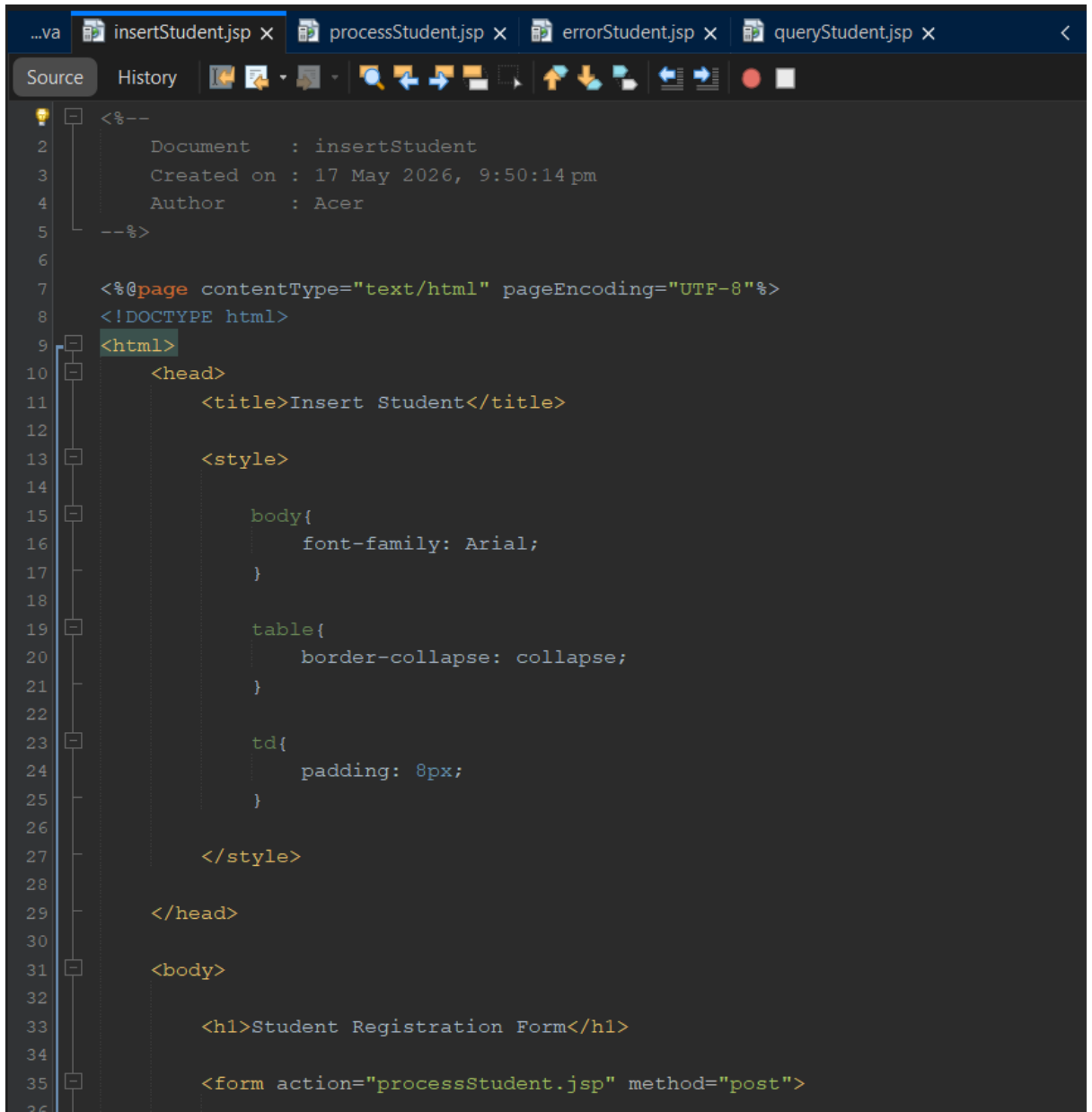
Program ▼

Address

Error Occurred!

Student ID must follow format like UK12345

[Back to Student Form](#)



```
...va insertStudent.jsp x processStudent.jsp x errorStudent.jsp x queryStudent.jsp x
Source History
1 <%--
2     Document    : insertStudent
3     Created on  : 17 May 2026, 9:50:14 pm
4     Author     : Acer
5 --%>
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11     <title>Insert Student</title>
12
13     <style>
14
15         body{
16             font-family: Arial;
17         }
18
19         table{
20             border-collapse: collapse;
21         }
22
23         td{
24             padding: 8px;
25         }
26
27     </style>
28
29 </head>
30
31 <body>
32
33     <h1>Student Registration Form</h1>
34
35     <form action="processStudent.jsp" method="post">
36
```

```
<table>
  <tr>
    <td>Student ID</td>
    <td>
      <input type="text" name="stuid"
        placeholder="Example: UK12345" required> </td>
    </tr>
    <tr> <td>Student Name</td>
      <td>
        <input type="text" name="stuname" required></td>
      </tr>
    <tr><td>Program</td>
      <td>
        <select name="stuprogram">
          <option>BSc Soft. Eng.</option>
          <option>BSc with IM</option>
          <option>BSc in Robotics</option>
        </select>
      </td>
    </tr>
    <tr><td>Address</td>
      <td>
        <textarea name="address"></textarea>
      </td>
    </tr>
    <tr>
      <td colspan="2">
        <input type="submit" value="Submit">
        <input type="reset" value="Cancel">
      </td>
    </tr>
  </table>
</form>
</body>
</html>
```



```
<%--
    Document      : processStudent
    Created on    : 17 May 2026, 9:50:45 pm
    Author       : Acer
--%>
<%@page import="java.sql.*"%>
<%@page import="lab6.com.Student"%>

<!DOCTYPE html>
<html>
    <head>
        <title>Process Student</title>
    </head>

    <body>

        <h1>Process Student Registration</h1>

    <%

        Connection conn = null;
        PreparedStatement ps = null;

        try {

            Student student = new Student();

            student.setStuid(request.getParameter("stuid"));
            student.setStuname(request.getParameter("stuname"));
            student.setStuprogram(request.getParameter("stuprogram"));
            student.setAddress(request.getParameter("address"));

            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
Source History
conn = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/CSA3203",
    "root",
    "admin"
);

String sql
    = "INSERT INTO Student VALUES(?,?,?,?)";

ps = conn.prepareStatement(sql);

ps.setString(1, student.getStuid());
ps.setString(2, student.getStuname());
ps.setString(3, student.getStuprogram());
ps.setString(4, student.getAddress());

ps.executeUpdate();

%>

<h2>Student record inserted successfully!</h2>

<table border="1">

    <tr>
        <th>Student ID</th>
        <th>Name</th>
        <th>Program</th>
        <th>Address</th>
    </tr>

    <tr>
        <td><%= student.getStuid() %></td>
        <td><%= student.getStuname() %></td>
        <td><%= student.getStuprogram() %></td>
        <td><%= student.getAddress() %></td>
    </tr>

</table>
```

```

2      </table>
3
4
5      <%
6
7          } catch (Exception e) {
8
9              request.setAttribute("errorMessage", e.getMessage());
10
11              RequestDispatcher rd
12                  = request.getRequestDispatcher("errorStudent.jsp");
13
14              rd.forward(request, response);
15
16          } finally {
17
18              if (ps != null) {
19                  ps.close();
20              }
21
22              if (conn != null) {
23                  conn.close();
24              }
25          }
26      %>
27
28      </body>
29  </html>

```

```
<%
    } catch (Exception e) {

        request.setAttribute("errorMessage", e.getMessage());

        RequestDispatcher rd
            = request.getRequestDispatcher("errorStudent.jsp");

        rd.forward(request, response);

    } finally {

        if (ps != null) {
            ps.close();
        }

        if (conn != null) {
            conn.close();
        }

    }

%>

</body>
</html>
```

```

6      } catch (Exception e) {
7
8          request.setAttribute("errorMessage", e.getMessage());
9
10         RequestDispatcher rd
11             = request.getRequestDispatcher("errorStudent.jsp");
12
13         rd.forward(request, response);
14
15     } finally {
16
17         if (ps != null) {
18             ps.close();
19         }
20
21         if (conn != null) {
22             conn.close();
23         }
24     }
25
26     %>
27
28     </body>
29
30     </html>

```

```

9      request.setAttribute("errorMessage", e.getMessage());
10
11      RequestDispatcher rd
12          = request.getRequestDispatcher("errorStudent.jsp");
13
14      rd.forward(request, response);
15
16  } finally {
17
18      if (ps != null) {
19          ps.close();
20      }
21
22      if (conn != null) {
23          conn.close();
24      }
25  }
26
27  %>
28
29  </body>
30  </html>

```

```

1      RequestDispatcher rd
2          = request.getRequestDispatcher("errorStudent.jsp");
3
4      rd.forward(request, response);
5
6  } finally {
7
8      if (ps != null) {
9          ps.close();
10     }
11
12     if (conn != null) {
13         conn.close();
14     }
15 }
16 %>
17
18 </body>
19 </html>

```

```

2         = request.getRequestDispatcher("errorStudent.jsp");
3
4         rd.forward(request, response);
5
6     } finally {
7
8         if (ps != null) {
9             ps.close();
10        }
11
12        if (conn != null) {
13            conn.close();
14        }
15    }
16
17    %>
18
19    </body>
20    </html>

```

```

4         rd.forward(request, response);
5
6     } finally {
7
8         if (ps != null) {
9             ps.close();
10        }
11
12        if (conn != null) {
13            conn.close();
14        }
15    }
16
17    %>
18
19    </body>
20    </html>

```

```

6         } finally {
7
8             if (ps != null) {
9                 ps.close();
10            }
11
12            if (conn != null) {
13                conn.close();
14            }
15        }
16    }
17
18    %>
19
20    </body>
21    </html>

```

```

8         if (ps != null) {
9             ps.close();
10        }
11
12        if (conn != null) {
13            conn.close();
14        }
15    }
16    %>
17
18    </body>
19    </html>

```

```

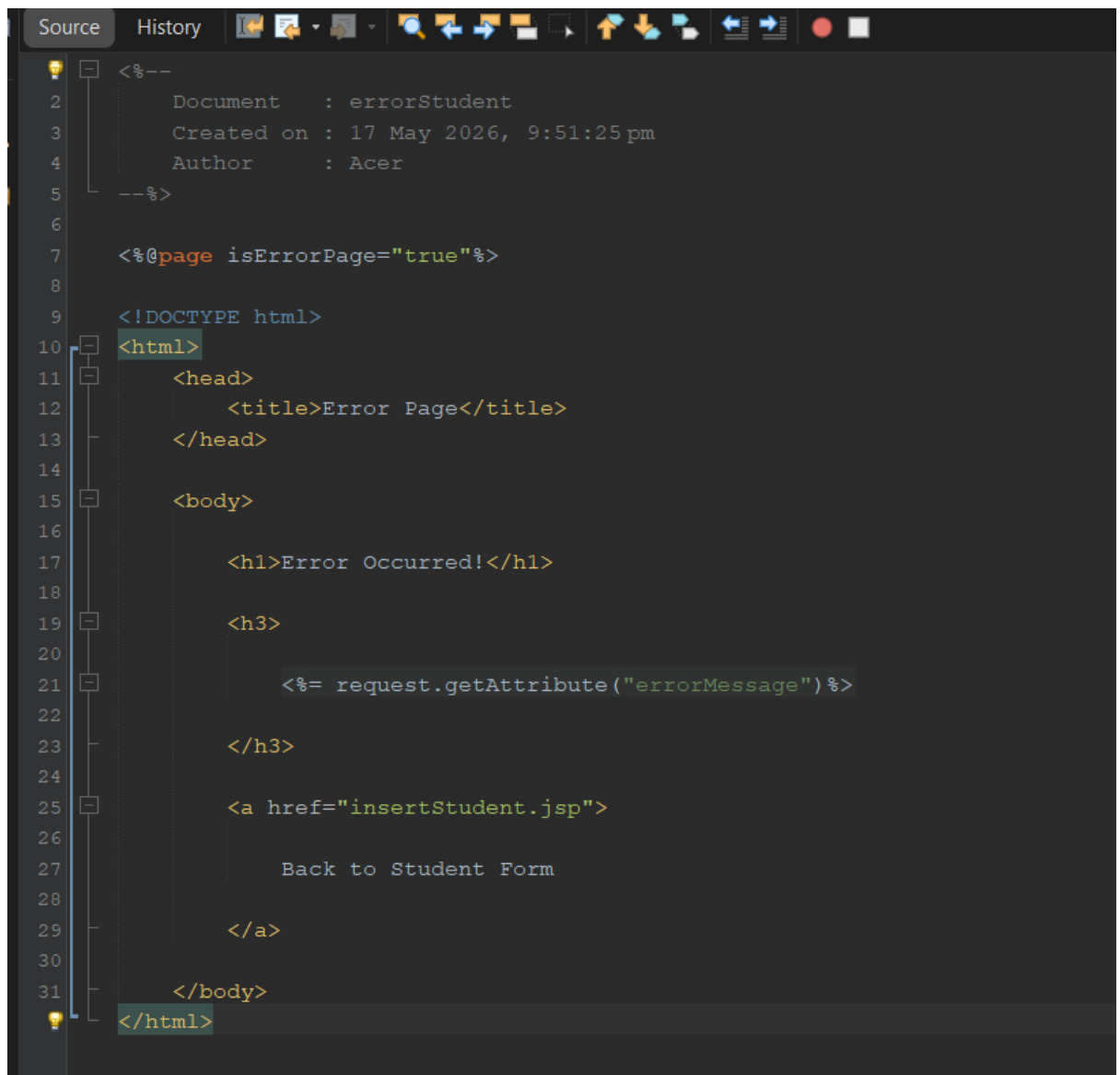
1         if (conn != null) {
2             conn.close();
3         }
4     }
5 }
6 %>
7
8 </body>
9 </html>

```

```
6 |         %>
7 |
8 |     </body>
9 | </html>
```

```
</body>
</html>
```

```
</html>
```



```
source History
Document : queryStudent
Created on : 17 May 2026, 9:51:57 pm
Author : Acer
--%>

<%@page import="java.sql.*"%>

<!DOCTYPE html>
<html>
  <head>
    <title>Query Student</title>

    <style>
      body{
        font-family: Arial;
      }
      table{
        width: 80%;
        border-collapse: collapse;
      }
      th, td{
        border: 1px solid black;
        padding: 10px;
        text-align: center;
      }
    </style>
  </head>
  <body>
    <h1>Lab 6 - Task 4: Retrieving record via JSP Page</h1>
    <table>
      <tr>
        <th>Student ID</th>
        <th>Student Name</th>
        <th>Program</th>
        <th>Address</th>
      </tr>
```

```
source History
0      Connection conn = null;
1      Statement stmt = null;
2      ResultSet rs = null;
3      try {
4          Class.forName("com.mysql.cj.jdbc.Driver");
5          conn = DriverManager.getConnection(
6              "jdbc:mysql://localhost:3306/CSA3203",
7              "root",
8              "admin");
9          stmt = conn.createStatement();
0      String sql = "SELECT * FROM Student";
1      rs = stmt.executeQuery(sql);
2      while (rs.next()) {
3          %>
4          <tr>
5              <td>
6                  <%= rs.getString("stuid")%>
7              </td>
8              <td>
9                  <%= rs.getString("stuname")%>
0              </td>
1              <td>
2                  <%= rs.getString("stuprogram")%>
3              </td>
4              <td>
5                  <%= rs.getString("address")%>
6              </td>
7          </tr>
8          <%
9              }
0          } catch (Exception e) {
1              out.println(e);
2          } finally {
3              if (rs != null) {
4                  rs.close();
5              }
```

```

73         if (rs != null) {
74             rs.close();
75
76             if (stmt != null) {
77                 stmt.close();
78             }
79
80             if (conn != null) {
81                 conn.close();
82             }
83         }
84     }
85 }
86
87 </html>

```

```

/**
 *
 * @author Acer
 */
package lab6.com;

public class Student {

    private String stuid;
    private String stuname;
    private String stuprogram;
    private String address;

    public Student() {
    }

    public String getStuid() {
        return stuid;
    }

    public void setStuid(String stuid) {
        if (stuid.matches("[A-Z]{2}[0-9]{5}")) {
            this.stuid = stuid;
        } else {
            throw new IllegalArgumentException(
                "Student ID must follow format like UK12345"
            );
        }
    }

    public String getStuname() {
        return stuname;
    }
}

```

```
public String getStuname() {  
    return stuname;  
}  
  
public void setStuname(String stuname) {  
    this.stuname = stuname;  
}  
  
public String getStuprogram() {  
    return stuprogram;  
}  
  
public void setStuprogram(String stuprogram) {  
    this.stuprogram = stuprogram;  
}  
  
public String getAddress() {  
    return address;  
}  
  
public void setAddress(String address) {  
    this.address = address;  
}  
}
```

Reflection

1. What have you learnt from this exercise?
2. Define step by step before you successfully perform the transaction in a database.

TASK 4

Student Registration Form

Student ID	UK56789
Student Name	Rosnah Azman
Program	BSc Soft. Eng. v
Address	
Submit Cancel	

Process Student Registration

Student record inserted successfully!

Student ID	Name	Program	Address
UK56789	Rosnah Azman	BSc Soft. Eng.	

Student Registration Form

Student ID	UK67342
Student Name	Liew Cheng Huat
Program	BSc in Robotics
Address	
Submit Cancel	

Process Student Registration

Student record inserted successfully!

Student ID	Name	Program	Address
UK67342	Liew Cheng Huat	BSc in Robotics	

Lab 6 - Task 4: Retrieving record via JSP Page

Student ID	Student Name	Program	Address
UK12489	Ahmad Salam	BSc with IM	
UK56789	Rosnah Azman	BSc Soft. Eng.	
UK67342	Liew Cheng Huat	BSc in Robotics	

Reflection

1. What have you learnt from this exercise?

I learned how to create a login system using JSP and MySQL database.

2. Explain the differences when using Statement() and PreparedStatement().

PreparedStatement is safer and helps prevent SQL Injection compared to Statement.

TASK 5

Run SQL query/queries on database csa3203: 

```
1 CREATE TABLE marathon (  
2     participantID VARCHAR(15) PRIMARY KEY,  
3     participantName VARCHAR(50),  
4     category VARCHAR(30),  
5     contactNo VARCHAR(20)  
6 );
```

Marathon Registration Result

Registration Failed!

```
1 <%--
2     Document    : processMarathon
3     Created on  : 17 May 2026, 10:16:13 pm
4     Author     : Acer
5 --%>
6 <%@page import="lab6.com.Marathon"%>
7 <%@page import="lab6.com.MarathonDAO"%>
8
9 <!DOCTYPE html>
10 <html>
11     <head>
12         <title>Process Marathon</title>
13     </head>
14
15     <body>
16
17         <h1>Marathon Registration Result</h1>
18
19         <%
20
21             Marathon m = new Marathon();
22
23             m.setParticipantID(
24                 request.getParameter("participantID")
25             );
26
27             m.setParticipantName(
28                 request.getParameter("participantName")
29             );
30
31             m.setCategory(
32                 request.getParameter("category")
33             );
```

```
Database.java X MarathonDAO.java X registerMarathon.jsp X processMarathon.jsp X
Source History
m.setContactNo(
    request.getParameter("contactNo")
);

MarathonDAO dao = new MarathonDAO();

boolean status = dao.addMarathon(m);

if (status) {

%>

<h2>Registration Successful!</h2>

<table border="1">

    <tr>
        <th>Participant ID</th>
        <td><%= m.getParticipantID() %></td>
    </tr>

    <tr>
        <th>Participant Name</th>
        <td><%= m.getParticipantName() %></td>
    </tr>

    <tr>
        <th>Category</th>
        <td><%= m.getCategory() %></td>
    </tr>

    <tr>
        <th>Contact Number</th>
        <td><%= m.getContactNo() %></td>
    </tr>

</table>
```

```
        </table>

    <%

    } else {

    %>

    <h2>Registration Failed!</h2>

    <%    }

    %>

</body>
</html>
```

```
<%--
    Document      : registerMarathon
    Created on    : 17 May 2026, 10:15:46 pm
    Author       : Acer
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>

        <title>Marathon Registration</title>

        <style>

            body{
                font-family: Arial;
            }

            table{
                border-collapse: collapse;
            }

            td{
                padding: 10px;
            }

        </style>

    </head>

    <body>

        <h1>Marathon Registration Form</h1>

        <form action="processMarathon.jsp" method="post">
```

```
<table>
  <tr>
    <td>Participant ID</td>
    <td>
      <input type="text" name="participantID" required>
    </td>
  </tr>
  <tr>
    <td>Participant Name</td>
    <td>
      <input type="text" name="participantName" required>
    </td>
  </tr>
  <tr>
    <td>Category</td>
    <td>
      <select name="category">
        <option>5 KM</option>
        <option>10 KM</option>
        <option>21 KM</option>
        <option>42 KM</option>
      </select>
    </td>
  </tr>

  <tr>
    <td>Contact Number</td>
    <td>
      <input type="text" name="contactNo">
    </td>
  </tr>
  <tr>
    <td colspan="2">
      <input type="submit" value="Register">
      <input type="reset" value="Cancel">
    </td>
  </tr>
```

```
    </td>

  </tr>

</table>

</form>

</body>
</html>
```



```

/**
 *
 * @author Acer
 */
package lab6.com;

import java.sql.Connection;
import java.sql.PreparedStatement;

public class MarathonDAO {

    public boolean addMarathon(Marathon m) {

        boolean status = false;

        Connection conn = null;
        PreparedStatement ps = null;

        try {

            conn = Database.getConnection();

            String sql
                = "INSERT INTO marathon VALUES(?,?,?,?)";

            ps = conn.prepareStatement(sql);

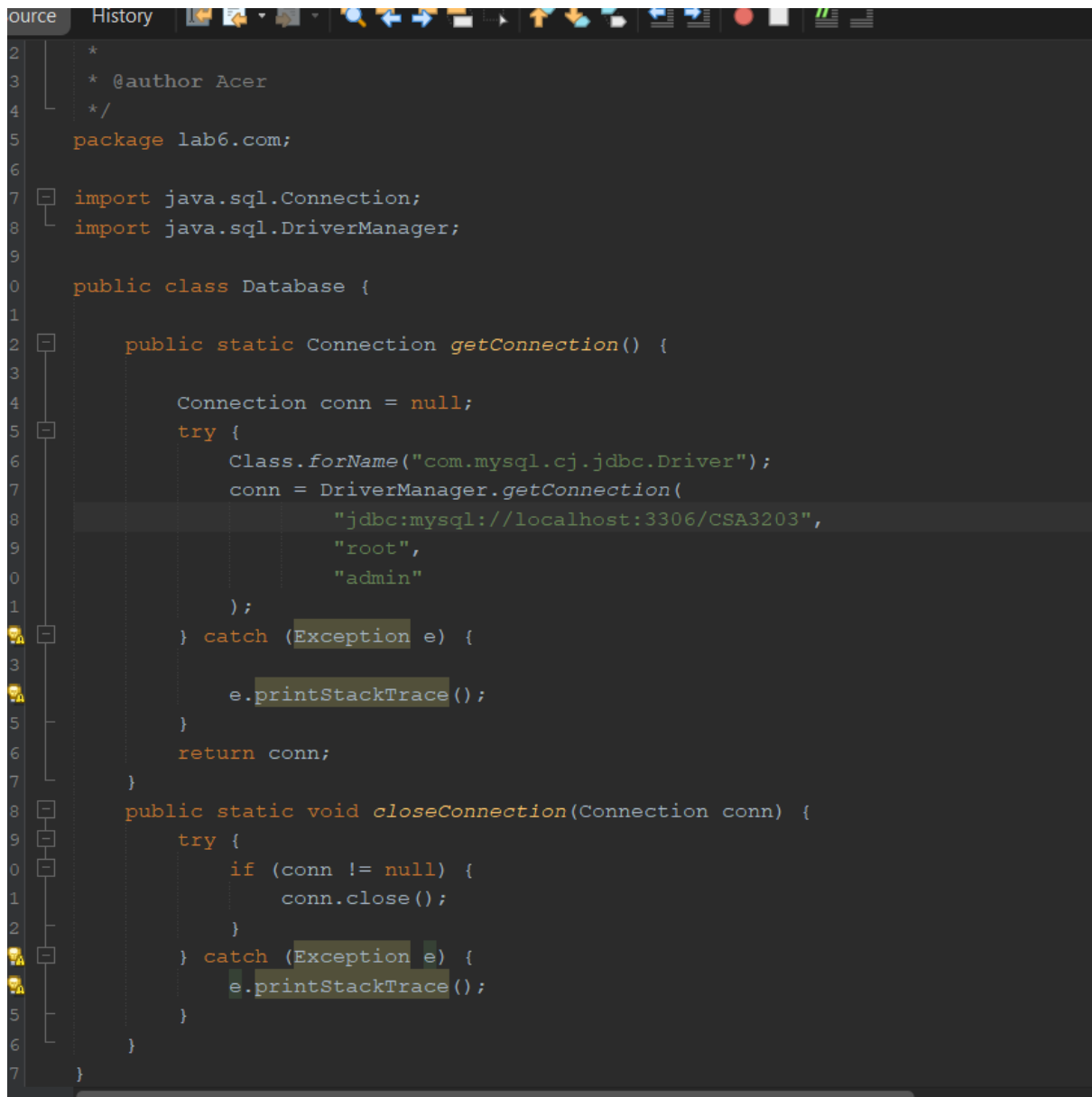
            ps.setString(1, m.getParticipantID());
            ps.setString(2, m.getParticipantName());
            ps.setString(3, m.getCategory());
            ps.setString(4, m.getContactNo());

            ps.executeUpdate();

            status = true;

```

```
    } catch (Exception e) {  
        e.printStackTrace();  
    } finally {  
        try {  
            if (ps != null) {  
                ps.close();  
            }  
  
            Database.closeConnection(conn);  
        } catch (Exception e) {  
            e.printStackTrace();  
        }  
    }  
  
    return status;  
}  
}
```

A screenshot of an IDE window showing a Java source file. The code defines a package 'lab6.com', imports 'java.sql.Connection' and 'java.sql.DriverManager', and defines a public class 'Database'. The class has two static methods: 'getConnection()' which attempts to establish a database connection using 'Class.forName()' and 'DriverManager.getConnection()', and 'closeConnection()' which closes the connection if it is not null. Both methods include try-catch blocks for 'Exception'. The IDE interface includes a toolbar at the top and a file explorer on the left.

```
2 | *
3 | * @author Acer
4 | */
5 | package lab6.com;
6 |
7 | import java.sql.Connection;
8 | import java.sql.DriverManager;
9 |
10 | public class Database {
11 |
12 |     public static Connection getConnection() {
13 |
14 |         Connection conn = null;
15 |         try {
16 |             Class.forName("com.mysql.cj.jdbc.Driver");
17 |             conn = DriverManager.getConnection(
18 |                 "jdbc:mysql://localhost:3306/CSA3203",
19 |                 "root",
20 |                 "admin"
21 |             );
22 |         } catch (Exception e) {
23 |             e.printStackTrace();
24 |         }
25 |         return conn;
26 |     }
27 |
28 |     public static void closeConnection(Connection conn) {
29 |         try {
30 |             if (conn != null) {
31 |                 conn.close();
32 |             }
33 |         } catch (Exception e) {
34 |             e.printStackTrace();
35 |         }
36 |     }
37 | }
```

Reflection

1. What have you learnt from this exercise?

I learned how to use JavaBeans and DAO classes to separate business logic and database operations from JSP pages.

2. Describe the benefits of using JavaBeans.

- Reusable code
- Easier maintenance
- Better organization
- Improves readability

EXERCISE

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0479 seconds.)

```
CREATE TABLE userprofile ( username VARCHAR(15) PRIMARY KEY, password VARCHAR(10), firstname VARCHAR(50), lastname VARCHAR(50) );
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

```
<!DOCTYPE html>
<html>
  <head>
    <title>User Registration</title>
    <style>
      body{
        font-family: Arial;
      }
      table{
        border-collapse: collapse;
      }
      td{
        padding: 10px;
      }
    </style>
  </head>
  <body>
    <h1>User Registration Form</h1>
    <form action="processUser.jsp" method="post">
      <table>
        <tr>
          <td>Username</td>
          <td>
            <input type="text" name="username" required>
          </td>
        </tr>
        <tr>
          <td>Password</td>
          <td>
            <input type="password" name="password" required>
          </td>
        </tr>
        <tr>
          <td>First Name</td>
          <td>
            <input type="text" name="firstname" required>
          </td>
        </tr>
      </table>
    </form>
  </body>
</html>
```

```
<td>First Name</td>
<td>
  <input type="text" name="firstname" required>
</td>
</tr>
<tr>
  <td>Last Name</td>
  <td>
    <input type="text" name="lastname" required>
  </td>
</tr>
<tr>
  <td colspan="2">
    <input type="submit" value="Register">
    <input type="reset" value="Cancel">
  </td>
</tr>
</table>
</form>
</body>
</html>
```



```

String lastname
    = request.getParameter("lastname");
Class.forName("com.mysql.cj.jdbc.Driver");
conn = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/CSA3203",
    "root",
    "admin"
);
String sql
    = "INSERT INTO userprofile VALUES(?, ?, ?, ?)";
ps = conn.prepareStatement(sql);
ps.setString(1, username);
ps.setString(2, password);
ps.setString(3, firstname);
ps.setString(4, lastname);
ps.executeUpdate();
%>
<h2>User registered successfully!</h2>
<a href="login.jsp">
    Go to Login Page
</a>
<%
    } catch (Exception e) {
        out.println(e);
    } finally {
        if (ps != null) {
            ps.close();
        }
        if (conn != null) {
            conn.close();
        }
    }
%>
</body>
</html>

```

```
Source History
<!--
  Document      : login
  Created on   : 17 May 2026, 10:37:33 pm
  Author      : Acer
-->

<!DOCTYPE html>
<html>

  <head>

    <title>Login</title>

    <style>

      body{
        font-family: Arial;
      }

      table{
        border-collapse: collapse;
      }

      td{
        padding: 10px;
      }

      .error{
        color: red;
      }

    </style>

  </head>

  <body>
```



```

<body>
  <h1>Login System</h1>

  <%
    String msg
      = request.getParameter("msg");

    if (msg != null) {
  %>
  <p class="error">
    <%= msg%>
  </p>
  <%
    }
  %>

  <form action="doLogin.jsp" method="post">
    <table>
      <tr>
        <td>Username</td>
        <td>
          <input type="text" name="username">
        </td>
      </tr>
      <tr>
        <td>Password</td>
        <td>
          <input type="password" name="password">
        </td>
      </tr>
      <tr>
        <td colspan="2">
          <input type="submit" value="Login">
          <input type="reset" value="Cancel">
        </td>
      </tr>

```

```

    </tr>

    </table>

  </form>

</body>
</html>

```

```
<%--
    Document    : doLogin
    Created on  : 17 May 2026, 10:39:20 pm
    Author     : Acer
--%>
```

```
<%@page import="java.sql.*"%>
```

```
<%
    Connection conn = null;
    PreparedStatement ps = null;
    ResultSet rs = null;

    try {

        String username
            = request.getParameter("username");

        String password
            = request.getParameter("password");

        Class.forName("com.mysql.cj.jdbc.Driver");

        conn = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/CSA3023",
            "root",
            ""
        );

        String sql
            = "SELECT * FROM userprofile WHERE username=? AND password=?";

        ps = conn.prepareStatement(sql);
```

History

```
ps = conn.prepareStatement(sql);

ps.setString(1, username);
ps.setString(2, password);

rs = ps.executeQuery();

if (rs.next()) {

    String firstname
        = rs.getString("firstname");

    String lastname
        = rs.getString("lastname");

    response.sendRedirect(
        "main.jsp?username=" + username
        + "&firstname=" + firstname
        + "&lastname=" + lastname
    );

} else {

    response.sendRedirect(
        "login.jsp?msg=Invalid username or password..!"
    );

}

} catch (Exception e) {

    out.println(e);

} finally {

    if (rs != null) {
        rs.close();
    }
}
```

80:3

INS



Q Search



```
} catch (Exception e) {  
  
    out.println(e);  
  
} finally {  
  
    if (rs != null) {  
        rs.close();  
    }  
  
    if (ps != null) {  
        ps.close();  
    }  
  
    if (conn != null) {  
        conn.close();  
    }  
  
}
```

```
Source History
<%--
2   Document    : main
3   Created on  : 17 May 2026, 10:40:16 pm
4   Author     : Acer
5   --%>
6
7   <!DOCTYPE html>
8   <html>
9
10  <head>
11
12    <title>Main Page</title>
13
14    <style>
15
16      body{
17        font-family: Arial;
18      }
19
20    </style>
21
22  </head>
23
24  <body>
25
26    <h1>Welcome to the System</h1>
27
28    <h3>
29
30      Username:
31      <%= request.getParameter("username") %>
32
33    </h3>
34
35    <h3>
```

```
<h3>
    First Name:
    <%= request.getParameter("firstname") %>
</h3>

<h3>
    Last Name:
    <%= request.getParameter("lastname") %>
</h3>

</body>
</html>
```

User Registration Form

Username	izza
Password	
First Name	izza
Last Name	Hazira
Register Cancel	

User Registration Result

User registered successfully!

[Go to Login Page](#)

Login System

Username

Password

Login

Cancel

Welcome to the System

Username: izza

First Name: Izza

Last Name: Hazira

Login System

Username

Password

Login

Cancel

Login System

Invalid username or password..!

Username

Password

Login

Cancel

Reflection

1. What have you learnt from this exercise?

I learned how to create a login system using JSP and MySQL database.

2. Describe the benefits of using JavaBeans.

PreparedStatement is safer and helps prevent SQL Injection compared to Statement.